

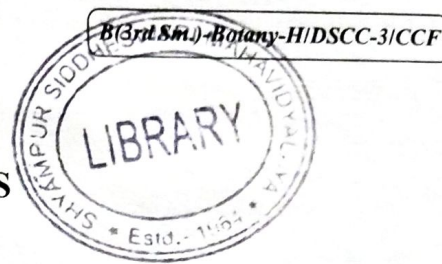
2024

BOTANY — HONOURS

Paper : DSCC-3

(Economic Botany)

Full Marks : 75



The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer **any six** questions from the following :

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|---|-----|
| (a) Distinguish between primary and secondary centre of origin. | 2 |
| (b) What is green manure? Give one example. | 1+1 |
| (c) Write the scientific names of the plants yielding – ceara rubber and landolphia rubber. | 1+1 |
| (d) Write the scientific name and parts used to produce oil from mustard and sunflower. | 1+1 |
| (e) Mention the scientific name and active constituent of poppy plant. | 1+1 |
| (f) What are the differences between spices and condiments? | 2 |
| (g) Mention two uses of sissoo wood. | 2 |
| (h) State the sources of fibres from cotton and flax plant. | 1+1 |
| (i) Mention the types of fruits – tamarind and custard apple. | 1+1 |

2. Answer **any three** of the following :

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|--|-----------|
| (a) What do you mean by narcotics? Mention the health hazards of <i>Cannabis</i> plant. | 2+3 |
| (b) Write a brief note on the methods of propagation of potato. | 5 |
| (c) Write the scientific name, parts used and uses of any two of the following :
Coriander, Black pepper, Clove, Cinnamon. | (1+½+1)×2 |
| (d) Describe in brief the extraction process of jute fibres. | 5 |
| (e) Write the scientific names of Jowar and Bajra. Comment on the significance of millets as future cereals. | 2+3 |

3. Answer **any four** of the following :

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|---|-------|
| (a) What is 'centre of origin of cultivated crops'? Briefly describe the Vavilovian centre of origin of cultivated crop species mentioning the specific crops of each region. Comment on the reasons behind the loss of genetic diversity of crops. | 2+6+4 |
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- (b) Describe the process of rubber tapping and explain the steps involved in the processing of natural rubber from Para-rubber (*Hevea brasiliensis*) and Assam rubber (*Ficus elastica*).
6+6
- (c) Distinguish between essential oils and fats. Discuss, in brief, different extraction methods of essential oils. Write the scientific name, family and uses of groundnut and coconut.
 $2+6+[(1+\frac{1}{2}+\frac{1}{2})\times 2]$
- (d) Comment on different types of teas based on their processing method. Enumerate briefly the steps involved in processing of black tea. What are the uses of tea and coffee? 3+6+3
- (e) Describe the steps involved in the processing of sugarcane. Name the products and by-products of sugarcane industry. How is sugar extracted from sugar beet? 7+2+3
- (f) Write the scientific name, family and edible parts of the following : $(1+\frac{1}{2}+\frac{1}{2})\times 6$
- (i) Cauliflower
 - (ii) Tomato
 - (iii) Sweet potato
 - (iv) Pineapple
 - (v) Pomegranate
 - (vi) Litchi.
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